Capability-Structured Agentic Control for Joint Conditional Handover and PDCP Duplication in LEO Satellite Networks

Jiwoong Jeon and Hoki Baek, *Member, IEEE*

***Abstract*—In low Earth orbit (LEO) satellite networks, user equipment (UE) must change its serving satellite frequently. Conditional handover (CHO) and packet data convergence protocol (PDCP) duplication can protect service continuity, but they turn mobility into coupled preparation, duplication, and switching decisions while UEs compete for shared satellite resources under delayed and noisy telemetry. We formulate the problem as a constrained decentralized partially observable Markov decision process and develop a centralized-training, decentralized-execution framework in which recurrent memory, ephemeris-based lookahead, congestion-price coordination, and primal–dual reliability regulation are introduced as separate agent capabilities. The control loop does not invoke a language model. All 25 independently trained policies achieve a lower outage ratio than a tuned 3GPP-inspired CHO rule in every evaluation scenario, and the complete model reduces the outage ratio by 60% under nominal conditions and by 46% under compound stress. Compared with a memoryless learned policy, it attains a statistically indistinguishable outage ratio with 39% fewer handover executions and a 70% lower ping-pong ratio. A tuned multi-agent deep Q-network reaches a comparable outage ratio, whereas the complete model yields the lowest ping-pong ratio and lower across-seed variability. Ablations show that goal regulation produces the largest isolated reduction in handover executions.**

***Index Terms*—Agentic AI, conditional handover, PDCP duplication, LEO satellite networks, multi-agent reinforcement learning.**

I. Introduction

Non-terrestrial networks (NTNs) are expected to extend sixth-generation (6G) connectivity beyond terrestrial infrastructure and support ubiquitous service availability [1]–[3]. Among NTN platforms, low Earth orbit (LEO) constellations are particularly attractive because their lower altitude reduces propagation delay and path loss compared with geostationary systems while retaining wide-area coverage [1], [4]. However, orbital motion continuously changes satellite visibility and limits the time for which a satellite can serve a user equipment (UE), forcing even stationary UEs to hand over frequently [4], [5]. Consequently, LEO access requires mobility control that is proactive rather than purely reactive.

Proactive LEO mobility has therefore been studied from several perspectives. Classical, graph-, and flow-based methods select serving satellites using remaining visibility time, elevation, link quality, or load [4], [6]. Location-aware conditional handover (CHO) and AcceHo exploit predictable satellite motion to prepare or accelerate handover [7], [8], while SKYCASTLE reduces network-layer interruption through distributed satellite anchors [9]. PMPS instead replaces discrete handover interruption with predictive multi-path packet scheduling when a terminal can track several satellites simultaneously [10]. Together, these studies exploit mobility prediction at the target, path, or network layer, but they do not jointly control the protocol states that precede and follow a handover.

Those states arise directly from standardized mobility mechanisms. Multi-connectivity exploits multiple radio paths for redundancy and reliability [11]–[13]. Within New Radio (NR), conditional reconfiguration allows a candidate target to be prepared before the serving link deteriorates and executed later [14], whereas PDCP duplication can temporarily send the same packet stream over two lower-layer legs with explicit activation and deactivation [15]. Together, these mechanisms allow a UE to prepare a candidate, protect traffic with a second leg, switch when appropriate, and release unnecessary resources. A handover is therefore no longer a single trigger or target decision, but a sequence of coupled preparation, duplication, and switching decisions.

Managing this sequence is difficult because each action changes both future feasible actions and shared-resource usage. Aggressive preparation occupies scarce control and radio resources, persistent duplication wastes capacity, and frequent switching increases handover and ping-pong events, whereas late action exposes the UE to outage as the serving satellite moves away. The decisions of different UEs are also coupled because they compete for the same satellite capacity and preparation resources. Moreover, each UE observes link quality and load only through delayed and noisy telemetry, so the instantaneous observation may not reveal whether degradation is transient or persistent. We therefore treat mobility management as a sequential control problem under partial observability and shared-resource coupling.

Learning-based satellite mobility control has progressed from deep Q-network (DQN)-based handover decisions [16] to multi-agent schemes that account for distributed users and satellite load [17], [18]. GEMS embeds future satellite mobility into dual-connectivity configuration [19], while hierarchical reinforcement learning (RL) separates when to hand over from which satellite to select [20]. These methods adapt handover decisions to dynamic conditions, but their action spaces re-

(Corresponding author: Hoki Baek.)

The authors are with the School of Computer Science and Engineering, Kyungpook National University, Daegu 37224, South Korea (e-mail: jjw000628@knu.ac.kr; neloyou@knu.ac.kr).

main centered on handover timing or serving-target selection. Joint CHO/PDCP control additionally requires the policy to track protocol state and coordinate preparation, duplication, switching, release, and rollback over shared resources. This motivates a controller with memory, prediction, coordination, and explicit reliability regulation.

These requirements correspond to capabilities commonly associated with Agentic AI, including perception, memory, reasoning, goal-directed action, and multi-agent coordination [21]–[23]. Advanced deep RL (DRL) provides a practical implementation route for such sequential agents [24]. Recent wireless and satellite designs span large language model (LLM)-aided spectrum sharing, decentralized satellite-resource agents, task-oriented orchestration, hierarchical LLM–algorithm control, semantic space–air–ground integrated network (SAGIN) management, LLM-empowered MAPPO, and agentic multipath transmission [25]–[31]. These works illustrate both LLM-centric and domain-specific realizations of agentic control. For the mobility problem considered here, telemetry, ephemeris, and protocol states already provide structured numerical inputs; an LLM therefore need not be invoked at each control epoch.

Based on this observation, we develop a multi-agent framework for joint CHO and PDCP-duplication control in shared-capacity LEO networks. Each UE acts as an autonomous agent that prepares candidates, activates or deactivates duplication, commits a switch, releases a candidate, or rolls back, and all agents are trained using multi-agent proximal policy optimization (MAPPO) with centralized training and decentralized execution (CTDE). Recurrent memory summarizes local history, an ephemeris-based planner anticipates candidate viability, a scalar congestion price conveys shared-resource pressure without UE-to-UE state/action exchange, and a primal–dual regulator adapts reliability pressure during training. The resulting M1–M5 model family allows each added capability to be evaluated separately.

The main contributions of this work are summarized as follows.

- We formulate joint CHO preparation, PDCP-duplication control, and primary-satellite switching in a shared-capacity LEO network as a constrained decentralized partially observable Markov decision process. The formulation captures delayed/noisy telemetry, time-varying visibility, hidden link degradation, preparation contention, duplication overhead, and mobility cost.
- We develop a capability-structured CTDE-MAPPO framework that introduces recurrent memory, ephemeris-based reasoning, and resource-aware coordination as distinct capabilities. This structure enables capability-wise comparisons and targeted within-policy ablations without relying on a monolithic LLM controller.
- We introduce satellite congestion prices for lightweight decentralized coordination without UE-to-UE state/action exchange and a primal–dual reliability regulator that automatically adapts the outage penalty instead of using a fixed reliability weight.
- We evaluate independently trained capability variants against conventional handover/duplication heuristics, a 3GPP-inspired CHO baseline, and value- and policy-based multi-agent RL baselines under nominal and compound traffic, capacity, and blockage disturbances, together with targeted capability ablations.

II. Related Work

A. LEO Mobility Management and Conditional Handover

Satellite motion produces time-varying coverage, short serving intervals, and frequent handovers [5], so conventional schemes use satellite-specific information such as elevation, remaining visibility time, channel availability, and load. Zhang *et al.* formulate multi-user handover as a network-flow problem whose edge weights reflect user requests and satellite service quality [6], and Juan *et al.* analyze 5G NR handover over LEO networks using predictable satellite motion and antenna information [4]. Proactive mechanisms further exploit orbital predictability: Lee *et al.* trigger CHO from the distance to serving and target cell reference locations [7], AcceHo predicts future satellite access to reduce core-network interaction and handover latency [8], and SKYCASTLE uses distributed satellite anchors to reduce interruption at the network layer [9]. These approaches address when and where a UE hands over, but do not jointly determine candidate preparation, temporary PDCP duplication, and the subsequent switch.

B. Multi-Connectivity and PDCP Duplication

Multi-connectivity (MC) uses radio resources from multiple access points for aggregation or path redundancy [11], and simultaneous terrestrial 5G and Starlink connectivity has been shown to reduce high-end latency and improve service availability [12]. 3GPP considers MC and service continuity for NTN [13], conditional reconfiguration stores candidate configurations for later execution [14], and PDCP supports duplication with explicit activation and deactivation [15], but these specifications do not prescribe a joint policy for when to prepare, duplicate, switch, and release. GEMS formulates terrestrial–NTN and NTN–NTN dual connectivity as a temporally extended matching problem that accounts for future satellite mobility [19], whereas PMPS assumes terminals that track several satellites simultaneously and replaces the single-path handover by predictive multi-path packet scheduling with selective redundancy [10]. GEMS thus addresses connectivity association and PMPS continuous packet migration; our work instead retains the standardized CHO and PDCP mechanisms and jointly controls their lifecycle—candidate preparation, temporary duplication, switching, release, and rollback—under shared preparation and radio resources.

C. Learning-Based Satellite Mobility Control

RL has been applied to adaptive satellite mobility management. DQN-ALrM combines multiple mobility-related metrics in a DQN-based handover policy [16], while multi-agent DRL accounts for dynamic propagation, user-rate requirements, and satellite load [17]. Yang *et al.* formulate handover as a decentralized decision process and use multi-agent fingerprints with double DQN (DDQN) to mitigate non-stationarity and

load imbalance [18]. Zhao *et al.* instead separate handover timing and target selection using a recurrent PPO agent and a DQN agent [20]. These methods model handover primarily as a trigger or serving-target decision. A CHO/PDCP procedure additionally has an internal state in which preparation, duplication, switching, release, and rollback constrain future feasible actions and consume resources over multiple epochs; our framework explicitly models this stateful evolution under multi-user coupling and partial observability.

D. Agentic AI for Wireless and Satellite Networks

Agentic AI is commonly associated with contextual perception, persistent memory, reasoning, goal-directed behavior, autonomous action, and multi-agent coordination [21], [23]; the recent JSAC tutorial organizes future intelligent communications around closed-loop perception, reasoning, planning, tools, and memory [22], and advanced DRL has been identified as a key mechanism for sequential agent decisions [24]. LLM-centric designs use language models to interpret quality-of-service (QoS) or operator intent, to reason at a high level, or to guide lower-level learning, e.g., for LLM-aided spectrum sharing in LEO networks [25], in the Think-Do separation of LLM reasoning from real-time execution [28], in a monitor–analyze–plan–execute (MAPE-K) SAGIN architecture in which LLM agents cooperate with an RL learner [29], and in LLM-guided MAPPO for satellite–terrestrial power and rate allocation [30]. Domain-specific designs realize agentic behavior without an LLM in every control decision: AERS uses decentralized beam agents for satellite resource scheduling [26], task-oriented agents orchestrate resources in satellite-assisted Internet of Vehicles (IoV) [27], and distributed agents construct intent-aware routes combined with multipath congestion control in LEO networks [31]. Our controller follows the domain-specific direction but addresses a different control layer: recurrent state retains temporal evidence, ephemeris lookahead evaluates future mobility, congestion prices summarize shared-resource pressure, and a dual variable regulates reliability during training. These capabilities jointly govern the CHO/PDCP mobility procedure rather than spectrum, beam, routing, or power allocation, and the execution-time control remains numerical without an LLM.

III. System Model

We consider the downlink of an NTN in which a set $\mathcal{L} = \{1, \dots, L\}$ of LEO satellites serves a set $\mathcal{N} = \{1, \dots, N\}$ of ground UEs. Time is slotted into control epochs $t \in \{0, 1, \dots, T-1\}$ of duration Δ, and mobility decisions are taken once per epoch. Each UE is an autonomous decision-making agent that controls its own CHO and PDCP-duplication procedure, while all UEs share the finite radio and control-plane resources of the satellites. The main notations are summarized in Table I and the parameter values in Table II. Throughout the paper, calligraphic letters denote sets and bold letters denote vectors; a tilde marks a quantity observed by the agents through delayed and noisy telemetry, a hat marks a quantity predicted by the analytical planner of Section V, and an overbar marks an average.

TABLE I
Main notations

Symbol	Description
$\mathcal{N}$, N; $\mathcal{L}$, L	Set and number of UEs; of satellites
$\mathcal{K}_i(t)$, K	Candidate set of UE i and its size
$\mathcal{A}$, $a_i(t)$	Action set and action of UE i
t, Δ, T	Epoch index, epoch duration, episode length
$\ell_i(t)$, $c_i(t)$	Serving and target satellite of UE i
$\gamma_{i\ell}$, $e_{i\ell}$, $q_{i\ell}$	SINR (dB), spectral efficiency, packet success prob.
$\epsilon_{i\ell}$, $d_{i\ell}$, $v_{i\ell}$	Elevation, slant range, visibility indicator
λ_i, D_i, Q_i	Offered traffic rate (bit/s), demand, queue length
$x_{i\ell}$, $R_{i\ell}$, R_i	Requested resource fraction, leg rate, UE rate
κ_ℓ, ϖ	Capacity scale, reservation fraction per preparation
n_ℓ, m_ℓ, M	Reservations, new preparation requests, preparation slots
Φ_ℓ, U_ℓ	Requested pressure, aggregate intended demand
δ_i, u_i, α_i	Service deficit, outage indicator, handover success prob.
C_{def}, C_{ctrl}, C_{out}; C_{thr}	Deficit, control, outage costs; outage threshold
r, r_μ	Team reward, Lagrangian reward
$\mathbf{o}_i$, $\mathbf{b}_i$	Local observation, action mask
$\mathbf{g}_i$, $\boldsymbol{\zeta}_i$	Recurrent memory, experienced outcome
$\boldsymbol{\phi}_i$, $J_i(a)$	Reasoning feature, plan cost
y_ℓ, μ; π_θ, V_ψ	Congestion price, reliability multiplier; actor, critic

Modeling scope. The CHO procedure is represented by a system-level state machine that abstracts preparation, execution, completion, and cooldown, and PDCP duplication by the synchronized transmission of the same packet stream over the serving and the prepared target legs, following 3GPP TS 38.331 and TS 38.323 [14], [15]. Individual RRC messages, RLC entities, HARQ processes, and radio-link-failure recovery are not modeled, and the radio scheduler is fixed, so that the agents control only preparation, duplication, and switching.

Satellite ℓ moves on a circular orbit of altitude h and inclination ι. Its Earth-centered position is

$$\mathbf{p}_\ell(t) = (E+h)\big[\cos\Omega_\ell(t),\, \sin\Omega_\ell(t)\cos\iota,\, \sin\Omega_\ell(t)\sin\iota\big]^{\mathsf{T}}, \tag{1}$$

where E is the Earth radius and $\Omega_\ell(t) = \Omega_\ell(0) \pm 2\pi t\Delta/T_{\mathrm{orb}}$ is the orbital phase for the orbital period T_{orb}. The initial phases $\Omega_\ell(0)$ are evenly spaced over $[-0.38, 0.38]$ rad, and the sign is positive for $\Omega_\ell(0) \le 0$ and negative otherwise. This construction represents a stylized local constellation slice with overlapping satellite passes rather than a full Walker constellation. It creates repeated rise–set transitions and simultaneous handover opportunities while keeping the mobility-control experiment computationally tractable. The UEs are fixed on the Earth surface, with latitudes and longitudes drawn uniformly within a ± 0.04 rad $\times$ ± 0.05 rad area. The elevation angle $\epsilon_{i\ell}(t)$ and the slant range $d_{i\ell}(t)$ follow from the positions of satellite ℓ and UE i, and satellite ℓ is visible to UE i if $v_{i\ell}(t) = \mathbb{1}\{\epsilon_{i\ell}(t) \ge \epsilon_{\min}\} = 1$.

The received power of the link (i, ℓ) in dBm is

$$P_{i\ell}(t) = P_{\mathrm{T}} + G_{\mathrm{T}} + G_{\mathrm{R}} - \mathrm{FSPL}\big(d_{i\ell}(t)\big) - A_{\mathrm{atm}} - A_{\mathrm{impl}} - A_{\mathrm{el}}\big(\epsilon_{i\ell}(t)\big) - \chi_{i\ell}(t) - B_{i\ell}(t) - Z_{i\ell}(t), \tag{2}$$

where P_{T} is the transmit power, G_{T} and G_{R} are the transmit and receive antenna gains, $\mathrm{FSPL}(d) = 20\log_{10} d + 20\log_{10} f - 147.55\,\mathrm{dB}$ is the free-space path loss for the distance d in meters and the carrier frequency f in hertz, A_{atm} and A_{impl} are the atmospheric and implementation losses,

and $A_{\mathrm{el}}(\epsilon) = 1.5/\max\{0.2, \sin\epsilon\}$ dB is the additional low-elevation loss. The shadowing $\chi_{i\ell}(t)$ is a first-order autoregressive (AR(1)) process with correlation coefficient ρ_χ and standard deviation σ_χ, $B_{i\ell}(t)$ is the extra loss caused by blockage events, and $Z_{i\ell}(t)$ is the loss of a hidden link-degradation regime described below. With the receiver noise power $P_{\mathrm{N}} = -174 + 10\log_{10} W + F$ dBm for the bandwidth W and the noise figure F, and a fixed aggregate interference floor P_{I}, the signal-to-interference-plus-noise ratio (SINR) of a visible link in dB is

$$\gamma_{i\ell}(t) = P_{i\ell}(t) - 10\log_{10}\big(10^{P_{\mathrm{N}}/10} + 10^{P_{\mathrm{I}}/10}\big). \tag{3}$$

The link is abstracted by a capped Shannon efficiency and a smooth packet-success model,

$$e_{i\ell}(t) = \min\big\{e_{\max},\ 0.75\log_2\big(1 + 10^{\gamma_{i\ell}(t)/10}\big)\big\}, \tag{4}$$

$$q_{i\ell}(t) = \Big(1 + \exp\big(-0.75\,(\gamma_{i\ell}(t) + 2)\big)\Big)^{-1} v_{i\ell}(t), \tag{5}$$

where $e_{\max}$ is the spectral-efficiency cap and $q_{i\ell}$ is the per-packet success probability, which equals 0.5 at -2 dB.

UE i has an offered traffic rate $\lambda_i(t)$ in bit/s, and the number of packets of S bits arriving in epoch t is Poisson distributed with mean $\lambda_i(t)\Delta/S$. The nominal rate of each UE is drawn uniformly within $\pm 35\%$ of the mean rate $\bar{\lambda}$, and it is multiplied by a burst factor during traffic-burst events. Packets wait in a finite queue of length $Q_i(t) \le Q_{\max}$ and are dropped when their age exceeds the deadline T_{dl}. A UE is active if $Q_i(t) > 0$, and its demand is

$$D_i(t) = \min\big\{\lambda_i(t),\ Q_i(t)S/\Delta\big\}. \tag{6}$$

Each UE has a serving satellite $\ell_i(t)$ and, during a mobility procedure, a target satellite $c_i(t)$. A UE requests a data leg on $\ell_i(t)$ and, when duplicating, a second leg on $c_i(t)$. The satellites run a fixed demand-proportional scheduler that is *not* controlled by the agents. For a leg (i,ℓ), the requested resource fraction is

$$x_{i\ell}(t) = \min\Big\{1,\ \frac{D_i(t)}{W\, e_{i\ell}(t)\,\kappa_\ell(t)}\Big\}, \tag{7}$$

where $\kappa_\ell(t)$ is the capacity scale of satellite ℓ; it equals the nominal value κ_0 except during capacity-drop events, in which it is reduced by a factor (and never falls below 0.05). We set $x_{i\ell}(t) = 0$ if UE i has no leg on satellite ℓ. Let $n_\ell(t)$ be the number of UEs holding a preparation reservation on satellite ℓ; each reservation blocks a fraction ϖ of the satellite resource, which leaves the fraction $1 - \varpi n_\ell(t)$ for data. The scheduler grants every request when the total request fits into this fraction and scales all requests proportionally otherwise, which yields the leg rate

$$R_{i\ell}(t) = W e_{i\ell}(t)\kappa_\ell(t)\, x_{i\ell}(t) \min\Big\{1,\ \frac{1 - \varpi n_\ell(t)}{\sum_{j\in\mathcal{N}} x_{j\ell}(t)}\Big\}. \tag{8}$$

The requested pressure of satellite ℓ is $\Phi_\ell(t) = \varpi n_\ell(t) + \sum_{i\in\mathcal{N}} x_{i\ell}(t)$. The scheduler treats all UEs equally and is kept outside the learned action space, so that performance differences stem from mobility decisions rather than scheduling.

Delivery. Without duplication, UE i transmits at the rate $R_i = R_{i\ell_i}$ with the success probability $q_i = q_{i\ell_i}$. With PDCP duplication, the same packets are sent on both legs, so that $R_i = \min\{R_{i\ell_i}, R_{ic_i}\}$ and

$$q_i = (1-\varkappa)\big[1-(1-q_{i\ell_i})(1-q_{ic_i})\big] + \varkappa\max\{q_{i\ell_i}, q_{ic_i}\}, \tag{9}$$

where $\varkappa \in [0,1]$ is a common-mode dependence parameter. The number of delivered packets is $Y_i(t) \sim \mathrm{Bin}\big(\min\{\lfloor R_i(t)\Delta/S\rfloor, Q_i(t)\},\, q_i(t)\big)$.

Service metrics. The normalized service deficit and the outage indicator of an active UE are

$$\delta_i(t) = \frac{\big[D_i(t) - \min\{R_i(t), Q_i(t)S/\Delta\}\, q_i(t)\big]^+}{D_i(t)},$$
$$u_i(t) = \mathbb{1}\big\{Y_i(t)S < \beta D_i(t)\Delta\big\}, \tag{10}$$

where β is the minimum acceptable service ratio. Note that δ_i is a smooth expected-value quantity used for learning, whereas u_i is the realized, packet-level reliability event.

Each UE runs the five-state mobility state machine of Fig. 1: SINGLE (S), PREPARED (P), DUPLICATING (D), SWITCHING (W), and COOLDOWN (C). The candidate set $\mathcal{K}_i(t)$ contains the K visible non-serving satellites with the highest *observed* SINR (13). The per-UE action set is

$$\mathcal{A} = \{\textsc{Stay},\ \textsc{Prep}_1, \ldots, \textsc{Prep}_K,\ \textsc{DupOn},$$
$$\textsc{DupOff},\ \textsc{Commit},\ \textsc{Release},\ \textsc{Rollback}\}, \tag{11}$$

which contains $K + 6$ actions. A state-dependent hard mask $\mathbf{b}_i(t) \in \{0,1\}^{K+6}$ enforces protocol and physical validity:

- In S, $\textsc{Prep}_k$ requests CHO preparation towards the k-th candidate. If admitted, the UE enters P with the preparation timer T_{prep}.
- In P (after the timer expires), DUPON activates duplication (P→D), COMMIT executes the CHO (P→W), and RELEASE cancels the preparation (P→S).
- In D, DUPOFF returns to P, COMMIT executes the CHO, and ROLLBACK cancels both the duplication and the preparation.
- COMMIT is feasible only if the target is visible and the dwell condition $t - t_i \ge T_{\mathrm{dwell}}$ holds, where t_i is the epoch of the last completed switch of UE i.

If the target becomes invisible in P or D, the preparation is lost and the UE returns to S. In W, the source satellite remains the serving data leg and the target reservation is retained until the switching outcome is resolved.

After T_{sw} epochs in W, the switch succeeds with the probability

$$\alpha_i(t) = \alpha_0\, q_{ic_i}(t)\, \exp\big(-[\Phi_{c_i}(t-1) - 1]^+\big), \tag{12}$$

where α_0 is the base success probability and $\Phi_{c_i}(t-1)$ is the most recently realized requested pressure of the target, since the outcome is resolved before the scheduler runs in epoch t.

On success, c_i becomes the new serving satellite and the UE enters C for T_{cd} epochs; otherwise it returns to S on the old serving satellite. A successful switch back to the previous serving satellite within T_{pp} epochs is counted as a ping-pong.

Preparation admission. CHO preparation consumes two separate resources of the target satellite: the data-plane reservation ϖ and one of M control-plane preparation slots. All

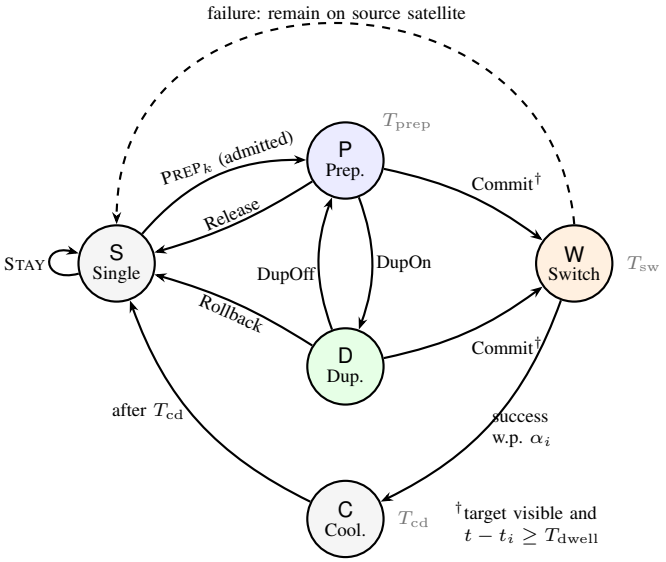


Fig. 1. Per-UE CHO and PDCP-duplication state machine. PREP$_k$ is subject to multi-UE admission; the timers T_{prep}, T_{sw}, and T_{cd} must elapse before leaving P, W, and C, respectively; a switch succeeds with the probability (12). Losing visibility of the target in P or D also returns the UE to S.

preparation requests of an epoch are collected first. Let $m_\ell(t)$ be the number of new requests towards satellite ℓ; if $m_\ell(t)$ exceeds $\min\{M - n_\ell(t),\ \lfloor(1 - \varpi n_\ell(t))/\varpi\rfloor\}$, the admitted requests are drawn uniformly at random and the others are rejected. Admission is therefore an explicit multi-UE contention.

The agents do not observe the complete physical state. Current geometric link availability is assumed to be known for the serving-link visibility indicator and candidate presence, whereas radio-quality, elevation, and load information provided to the actor is delayed by T_{obs} epochs. In addition, the observed SINR and load are corrupted by temporally correlated noise. Specifically,

$$\tilde{\gamma}_{i\ell}(t) = \gamma_{i\ell}(t - T_{\mathrm{obs}}) + \nu_{i\ell}(t), \tag{13}$$
$$\tilde{e}_{i\ell}(t) = \min\Big\{e_{\max}, \max\big\{0,\ e_{i\ell}(t - T_{\mathrm{obs}})\, 2^{\nu_{i\ell}(t)/10}\big\}\Big\},$$
$$\tilde{\epsilon}_{i\ell}(t) = \epsilon_{i\ell}(t - T_{\mathrm{obs}}),$$
$$\tilde{\Phi}_{i\ell}(t) = \big[\Phi_\ell(t - 1 - T_{\mathrm{obs}}) + \omega_{i\ell}(t)\big]^+,$$

where the telemetry noises $\nu_{i\ell}(t)$ and $\omega_{i\ell}(t)$ are independent zero-mean AR(1) processes with correlation coefficients ρ_ν and ρ_ω and standard deviations σ_ν and σ_ω, respectively. Thus, elevation telemetry is delayed but is not subjected to an additional measurement-noise term in the present implementation. The local observation $\mathbf{o}_i(t) \in \mathbb{R}^{16+7K}$ contains (i) the queue length, head-of-line delay, arrival rate, reliability-class indicator, duplication flag, one-hot protocol state, and protocol timer; (ii) for the serving satellite: visibility, $\tilde{\gamma}$, $\tilde{e}$, $\tilde{\epsilon}$, and $\tilde{\Phi}$; and (iii) for each candidate: presence, a prepared-target flag, $\tilde{\gamma}$, $\tilde{e}$, $\tilde{\epsilon}$, $\tilde{\Phi}$, and the observed SINR trend $\tilde{\gamma}(t) - \tilde{\gamma}(t-1)$. Agents do *not* observe the remaining visibility time, the capacity scales, the event realizations, the latent link regimes, or the states and actions of other UEs, and the action mask does not check whether a candidate remains visible long enough to complete a handover. The reliability-class indicator is fixed to 0.99 and is distinct from the outage threshold C_{thr} of Section IV.

Per-epoch control sequence. In each epoch, every UE selects one masked action from its observation, its protocol state, and its local history. The environment then resolves protocol timers and switching outcomes, replaces actions that have become infeasible by STAY, jointly admits the preparation requests, and serves the data legs, after which hidden regimes, events, telemetry, and congestion prices are updated. A price computed at the end of an epoch therefore affects only the next decision, so coordination is causal and one-epoch delayed.

Hidden link regimes. Every UE–satellite path (i, ℓ) follows a hidden semi-Markov regime GOOD $\rightarrow$ DEGRADED $\rightarrow$ RECOVERING $\rightarrow$ GOOD. A path enters DEGRADED with a small probability per epoch and with a larger probability at the start of an episode (Table II), stays there for a duration drawn uniformly between 8 and 24 epochs with an extra loss $Z_{i\ell}$ drawn uniformly between 7 and 15 dB, and then recovers over 4 to 10 epochs, during which $Z_{i\ell}$ decays linearly to zero. These regimes are observable only through their delayed, noisy effect on $\tilde{\gamma}$, which makes memory useful.

Disturbance events. Each training and nominal-evaluation episode contains five events whose start, duration, severity, affected satellite, and affected UE subset are sampled from the ranges in Table II: two blockages (extra loss $B_{i\ell}$ on the selected UE–satellite links), two capacity drops (reduced capacity scale κ_ℓ of one satellite), and one traffic burst (increased arrival rate λ_i of a subset of UEs). For the *compound-stress* evaluation, the randomized training-event schedule is replaced by a deterministic overlapping disturbance pattern: a traffic burst with the factor 2.5 on 50% of the UEs during $t \in [60, 120)$, a capacity drop of satellite 2 to 45% of its nominal capacity scale during $t \in [80, 130)$, and an 18-dB blockage affecting all links of 25% of the UEs during $t \in [100, 125)$. The burst and blockage severities lie outside their training ranges, the blockage affects all satellites of a UE, and the three disturbances overlap in time, which creates an operating condition not encountered during training.

IV. PROBLEM FORMULATION

A. Per-Epoch Costs

Let $\mathcal{N}(t) \subseteq \mathcal{N}$ be the set of UEs that are active in epoch t. We define the service, control, and reliability costs of epoch t as

$$C_{\mathrm{def}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} \delta_i(t), \tag{14}$$
$$C_{\mathrm{prep}}(t) = \frac{\varpi}{L} \sum_{\ell \in \mathcal{L}} n_\ell(t), \tag{15}$$
$$C_{\mathrm{HO}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} \mathbb{1}\{a_i(t) = \textsc{Commit}\}, \tag{16}$$
$$C_{\mathrm{ctrl}}(t) = C_{\mathrm{dup}}(t) + w_{\mathrm{prep}} C_{\mathrm{prep}}(t) + w_{\mathrm{HO}} C_{\mathrm{HO}}(t), \tag{17}$$
$$C_{\mathrm{out}}(t) = \frac{1}{|\mathcal{N}(t)|} \sum_{i \in \mathcal{N}(t)} u_i(t), \tag{18}$$

TABLE II
SIMULATION AND LEARNING PARAMETERS

Network and protocol	
Satellites / UEs / candidates	$L = 6$, $N = 12$, $K = 3$
Altitude, inclination, period	$h = 600\,\text{km}$, $\iota = 53^\circ$, $T_{\text{orb}} = 5800\,\text{s}$
Epoch, episode, min. elev.	$\Delta = 1\,\text{s}$, $T = 420$, $\epsilon_{\min} = 10^\circ$
Carrier, bandwidth	$f = 2\,\text{GHz}$, $W = 4\,\text{MHz}$
Tx power, Tx/Rx gain	$P_{\text{T}} = 46\,\text{dBm}$, $G_{\text{T}} = 30\,\text{dBi}$, $G_{\text{R}} = 0\,\text{dBi}$
Noise figure, interference	$F = 7\,\text{dB}$, $P_{\text{I}} = -108\,\text{dBm}$
Losses, shadowing	$A_{\text{atm}} = 0.5$, $A_{\text{impl}} = 3$, $\sigma_\chi = 2\,\text{dB}$; $\rho_\chi = 0.95$
SE cap, capacity scale	$e_{\max} = 6$, $\kappa_0 = 0.7$
Packet size, deadline, queue	$S = 12\,\text{kbit}$, $T_{\text{dl}} = 2\,\text{s}$, $Q_{\max} = 400$
Mean rate, spread	$\bar{\lambda} = 2\,\text{Mb/s}$, $\pm 35\%$
Min. service ratio, reliab. class	$\beta = 0.8$, 0.99 (all UEs)
$T_{\text{prep}}, T_{\text{sw}}, T_{\text{cd}}, T_{\text{dwell}}, T_{\text{pp}}$	$1, 2, 5, 15, 20$ epochs
Prep. reservation, control slots	$\varpi = 0.01$, $M = 2$
HO success, common-mode dep.	$\alpha_0 = 0.995$, $\varkappa = 0.15$
Partial observability and non-stationarity	
Telemetry lag	$T_{\text{obs}} = 2$ epochs
Telemetry noise	$\sigma_\nu = 1.5\,\text{dB}$, $\sigma_\omega = 0.08$, $\rho_\nu = \rho_\omega = 0.8$
Regime entry probability	0.06 (episode start), 0.006 (per epoch)
Blockage (2 events)	8–15 dB, 25–55 epochs, 30–65% of UEs
Capacity drop (2 events)	factor 0.45–0.80, 25–70 epochs
Traffic burst (1 event)	factor 1.35–1.85, 35–70 epochs
Objective and agentic modules	
Cost weights	$w = 0.25$, $w_{\text{prep}} = 0.2$, $w_{\text{HO}} = 0.05$
Outage threshold	$C_{\text{thr}} = 0.10$
Dual / price step size, bound	$\eta_\mu = 0.005$, $\mu_{\max} = 10$ / $\eta_y = 0.05$, $y_{\max} = 5$
Planning horizon, discount	$\tau_{\max} = 6$, $\xi_{\text{p}} = 0.95$
Learning (CTDE-MAPPO)	
Actor / critic / GRU	$[128, 128]$ / $[256, 256]$ / 128 (TBPTT 16)
Rollout, iterations, PPO epochs	$T_{\text{roll}} = 128$, $I = 2000$, 5 (batch 128)
Discount, GAE, clip, KL	$\xi = 0.99$, 0.95, 0.2, 0.03
LR actor / critic	$3{\cdot}10^{-4} \to 3{\cdot}10^{-5}$ / $10^{-3} \to 10^{-4}$

where the duplication cost $C_{\text{dup}}(t) = B_{\text{dup}}(t)/\big(B_{\text{use}}(t) + B_{\text{dup}}(t)\big)$ is the fraction of the scheduled bits that are duplicate copies B_{dup} on secondary legs, B_{use} being the bits scheduled on the serving legs. The control cost C_{ctrl} thus combines transmission redundancy, preparation reservations, and handover executions with the weights w_{prep} and w_{HO}; it does not charge preparation signaling messages (Section VIII).

B. Constrained Decentralized POMDP

The problem is a cooperative, constrained decentralized partially observable Markov decision process (Dec-POMDP) with the agent set $\mathcal{N}$, the state space $\mathcal{S}$, the action set $\mathcal{A}$ of every agent, the transition kernel $\mathcal{P}$, the team reward r, and the discount factor ξ. The state $s(t) \in \mathcal{S}$ comprises the satellite geometry, channel and shadowing states, hidden regimes and their timers, disturbance events and their remaining durations, the queues including packet ages, and the protocol states, timers, and last-switch epochs of all UEs; $\mathcal{P}$ is given by Section III; and each agent receives the local observation $\mathbf{o}_i(t)$ of Section III together with its action mask. All agents share the team reward

$$r(t) = -C_{\text{def}}(t) - w\, C_{\text{ctrl}}(t), \tag{19}$$

where w weights the control cost against the service deficit. Reliability is imposed as a constraint on the long-run outage cost rather than as a further reward weight:

$$\max_{\boldsymbol{\pi}} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \xi^t r(t)\Big] \ \text{ s.t. } \ \mathbb{E}_{\boldsymbol{\pi}}\big[C_{\text{out}}(t)\big] \le C_{\text{thr}}, \tag{20}$$

TABLE III
CAPABILITY-STRUCTURED MODEL FAMILY

Model	Capability added	Actor input $\mathbf{z}_i$	Training reward
M1	Perception (core)	$\mathbf{o}_i$	r
M2	+ Memory	$[\mathbf{o}_i, \mathbf{g}_i]$	r
M3	+ Reasoning	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i]$	r
M4	+ Coordination	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i, \mathbf{y}_i/y_{\max}]$	r
M5	+ Goal regulation	$[\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i, \mathbf{y}_i/y_{\max}, \mu/\mu_{\max}]$	r_μ

where C_{thr} is the outage threshold and $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_N)$ are decentralized policies that map the local observation history of each UE to its action. The per-satellite capacity constraints are enforced by the scheduler and the admission rule and need not be learned. We address Problem (20) through its Lagrangian relaxation

$$\max_{\boldsymbol{\pi}} \min_{\mu \ge 0} \ \mathbb{E}_{\boldsymbol{\pi}}\Big[\sum_t \xi^t r_\mu(t)\Big], \tag{21}$$
$$r_\mu(t) = r(t) - \mu\big(C_{\text{out}}(t) - C_{\text{thr}}\big),$$

where μ is the reliability multiplier; Section V-E describes the rollout-level primal–dual procedure used in practice.

Problem (20) is difficult because of (i) partial observability with delayed telemetry and hidden link regimes, (ii) delayed consequences, since a preparation blocks resources before any benefit appears and a target chosen now may set before the switch completes, and (iii) coupling, since UEs compete for shared resources and congestion lowers the handover success probability (12).

V. Proposed Capability-Structured Agentic Framework

We instantiate the above decomposition as five explicit control capabilities rather than as an LLM-based controller. M1 uses masked local observations, M2 adds recurrent memory, M3 adds ephemeris-based protocol lookahead, M4 adds a satellite congestion price, and M5 adds a training-time reliability multiplier. Table III summarizes the corresponding actor inputs and training objectives, and Fig. 2 shows their interaction. The M1–M5 sequence is used to isolate the effect of each added capability; it does not assume that performance improves monotonically. All variants use the same base reward except M5, which uses the reliability Lagrangian. Execution remains decentralized and numerical, with satellites broadcasting only scalar congestion prices.

A. Perception and Core Decision Making (M1)

All UEs share one stochastic actor π_θ with parameters θ and are trained with centralized training and decentralized execution (CTDE) using multi-agent proximal policy optimization (MAPPO) [32]. The actor maps its input $\mathbf{z}_i(t)$ to one logit per action through a neural network NN_θ, and invalid actions are removed by a masked softmax,

$$\pi_\theta(a \mid \mathbf{z}_i, \mathbf{b}_i) = \frac{b_{i,a} \exp\big([\text{NN}_\theta(\mathbf{z}_i)]_a\big)}{\mathbf{b}_i^{\mathsf{T}} \exp\big(\text{NN}_\theta(\mathbf{z}_i)\big)}, \tag{22}$$

where $b_{i,a}$ is the entry of $\mathbf{b}_i$ for action a and the exponential in the denominator is applied elementwise. Perception in M1

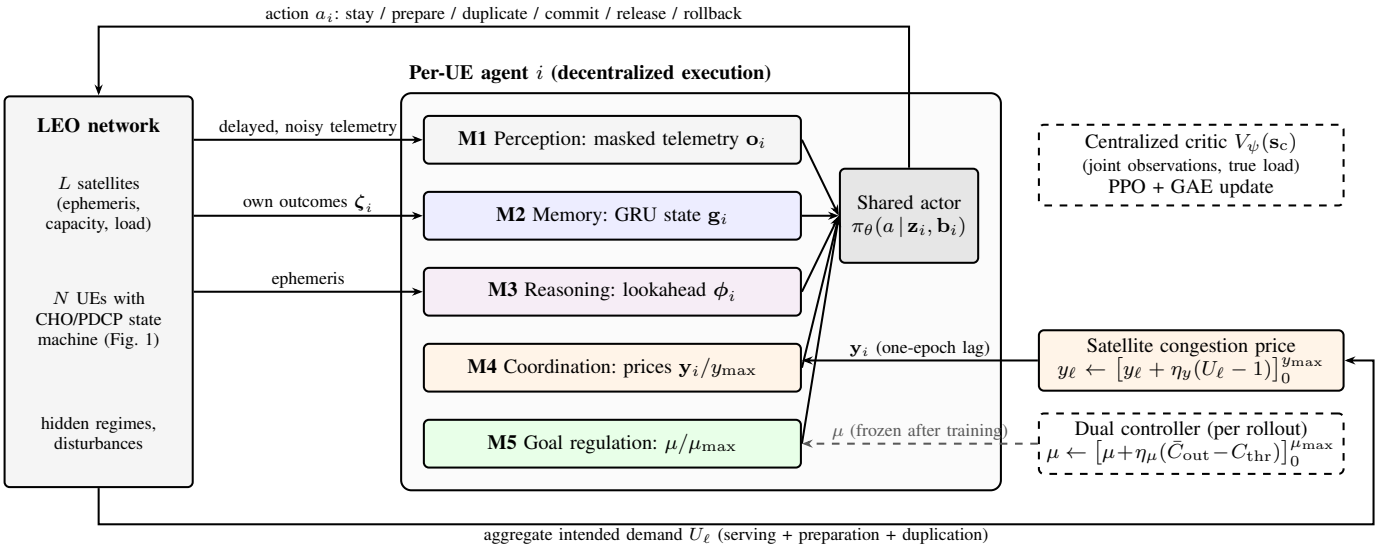


Fig. 2. Architecture of the capability-structured agent. Each UE runs the same actor on the concatenation $\mathbf{z}_i$ of the capability outputs available in its variant (M1: $\mathbf{o}_i$; M2 adds $\mathbf{g}_i$; M3 adds $\boldsymbol{\phi}_i$; M4 adds $\mathbf{y}_i/y_{\max}$; M5 adds $\mu/\mu_{\max}$). Solid: execution; dashed: training only. Satellites broadcast only the scalar prices.

is the masked, telemetry-based observation $\mathbf{z}_i = \mathbf{o}_i$. The centralized critic V_ψ with parameters ψ is used only during training. Its input $\mathbf{s}_c(t)$ concatenates the observations of all UEs with the true per-satellite utilization and pressure Φ_ℓ, and, when available, the satellite prices and the reliability multiplier.

B. Memory: Inferring Hidden Regimes (M2)

Because link regimes are hidden and telemetry is delayed, the current observation is not a sufficient statistic. M2 equips every agent with a gated recurrent unit (GRU) that summarizes its interaction history in the recurrent memory

$$\mathbf{g}_i(t) = \mathrm{GRU}_\theta\big(\mathbf{g}_i(t-1), [\mathbf{o}_i(t), \mathrm{onehot}(a_i(t-1)), \boldsymbol{\zeta}_i(t-1)]\big), \tag{23}$$

where $\boldsymbol{\zeta}_i \in \mathbb{R}^6$ is the agent's own *experienced outcome*: the last handover result ($+1$ success, -1 failure, 0 none), its outage indicator u_i, its deficit δ_i, the duplication gain (the part of its expected goodput contributed by the secondary leg, normalized by D_i), its resource-usage indicator, and the experienced congestion $\Phi/(1+\Phi)$ of its serving or target satellite. The actor input becomes $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{g}_i]$, so that the actor can condition its decision on recent local outcomes rather than on the current delayed observation alone. The GRU is trained end-to-end with truncated backpropagation through time.

C. Reasoning: Protocol-Aware Ephemeris Lookahead (M3)

The action mask enforces immediate feasibility but does not account for whether a target will remain visible long enough to complete the procedure, and the benefit of a preparation appears only several epochs later. M3 therefore adds an analytical lookahead based on the predictable satellite ephemeris. For every feasible first action a, the planner evaluates a small library $\mathcal{Q}_i(a)$ of protocol-consistent plans, i.e., action sequences $\mathbf{a} = (a(0), \ldots, a(\tau_{\max}-1))$ of length $\tau_{\max}$ that start with a (e.g., PREP$_k$–wait–COMMIT; PREP$_k$–wait–DUPON–COMMIT; STAY) and are padded with STAY, by rolling the UE's own protocol state machine forward. For tractability, this hypothetical rollout propagates protocol timers deterministically. Once a hypothetical COMMIT satisfies the dwell and predicted-visibility conditions, completion of the corresponding switch is treated as successful in the lookahead. Consequently, the stochastic handover-success model in (12) is realized by the environment but is not explicitly sampled or averaged inside the analytical planner.

The SINR of link (i,ℓ) at the lookahead step $\tau \in \{0, \ldots, \tau_{\max}-1\}$ is predicted from the *observed* SINR and the ephemeris-predicted range,

$$\hat{\gamma}_{i\ell}(t+\tau) = \tilde{\gamma}_{i\ell}(t) - 20\log_{10}\frac{d_{i\ell}(t+\tau)}{d_{i\ell}(t)}, \tag{24}$$

and the link is treated as unavailable once $\epsilon_{i\ell}(t+\tau) < \epsilon_{\min}$. This heuristic applies the ephemeris-predicted range change from t to $t+\tau$ to the delayed observation without correcting the observation delay. The predicted leg rate uses the nominal capacity scale κ_0 scaled by the observed load $\tilde{\Phi}_{i\ell}$, from which the planner obtains the predicted deficit $\hat{\delta}_i(t+\tau \mid \mathbf{a})$ under plan $\mathbf{a}$, including the duplication diversity of (9). The cost of a first action is the best discounted plan cost,

$$J_i(a) = \min_{\mathbf{a}\in\mathcal{Q}_i(a)} \sum_{\tau=0}^{\tau_{\max}-1} \xi_p^\tau \Big(\hat{\delta}_i(t+\tau \mid \mathbf{a}) + \mathbf{y}(t)^\mathsf{T}\big[\hat{\mathbf{x}}_i(t+\tau \mid \mathbf{a}) - \hat{\mathbf{x}}_i(t+\tau \mid \mathbf{a}_0)\big]^+ \Big), \tag{25}$$

where ξ_p is the planning discount, $\mathbf{a}_0$ is the plan that stays throughout, and the second term is present only in M4 and M5. Here, $\mathbf{y}(t) = [y_1(t), \ldots, y_L(t)]^\mathsf{T}$ is the raw price vector, and $\hat{\mathbf{x}}_i$ is the predicted per-satellite resource footprint of UE i (serving-leg fraction, secondary-leg fraction while duplicating, and the reservation ϖ while a target is prepared), so that only the incremental footprint relative to staying is priced. The

control-slot contention term m_ℓ/M of the aggregate demand U_ℓ (Section V-D) reaches the planner only through the price. The planner thus ranks plans by predicted service continuity, and in M4 and M5 additionally by recently observed shared-resource pressure; all other costs are learned through the team reward.

Let $J_{\min}$ and $J_{\max}$ be the minimum and maximum of $J_i(a)$ over the feasible actions of UE i. The reasoning feature of a feasible action is

$$\phi_{i,a} = \begin{cases} \dfrac{J_i(a) - J_{\min}}{J_{\max} - J_{\min}}, & J_{\max} - J_{\min} > 10^{-8}, \\ 0, & \text{otherwise}, \end{cases} \tag{26}$$

and $\phi_{i,a} = 1$ for masked actions. The vector $\boldsymbol{\phi}_i \in [0,1]^{K+6}$ is appended to the actor input: $\mathbf{z}_i = [\mathbf{o}_i, \mathbf{g}_i, \boldsymbol{\phi}_i]$. The planner is *advisory*: the actor learns how much to trust it. It has no access to future events, hidden regimes, true capacity scales, or the states of other UEs.

D. Coordination: Congestion Prices (M4)

UEs compete for preparation slots, reservations, and data capacity. To coordinate this coupling without direct UE-to-UE message exchange, M4 uses a scalar price maintained by each satellite. Let $\mathcal{D}(t)$ be the set of UEs that duplicate in epoch t. Satellite ℓ measures its aggregate intended demand

$$U_\ell(t) = \sum_{i:\, \ell_i(t)=\ell} x_{i\ell}(t) + \frac{m_\ell(t)}{M} + \sum_{i \in \mathcal{D}(t):\, c_i(t)=\ell} x_{i\ell}(t), \tag{27}$$

i.e., the serving demand, the new preparation requests normalized by the number of control slots, and the duplication demand, and updates its price y_ℓ by projected subgradient ascent with the step size η_y,

$$y_\ell(t+1) = \Big[y_\ell(t) + \eta_y \big(U_\ell(t) - 1 \big) \Big]_0^{y_{\max}}. \tag{28}$$

Prices rise while intended demand exceeds the normalized resource level and decay otherwise; they are reset at the start of each episode and are not learned. The price produced at the end of epoch t is used only for the next decentralized decision, so coordination is causal and one-epoch delayed.

The actor of UE i receives the normalized local price vector $\mathbf{y}_i(t)/y_{\max}$ with $\mathbf{y}_i(t) = [y_\ell(t)]_{\ell \in \{\ell_i(t)\} \cup \mathcal{K}_i(t)}$, whereas the planner uses the raw prices. The coordinator never allocates actions: the only coordination signal is one scalar per satellite and epoch, which reveals neither individual UE states nor actions and conveys only the recent aggregate resource pressure.

E. Goal Regulation: Primal–Dual Reliability Control (M5)

M5 addresses the Lagrangian problem (21). The dual update uses the realized outage fraction over all active-UE opportunities in one rollout. For the set $\mathcal{T}$ of the T_{roll} epochs of a rollout, the implementation computes the rollout-average outage cost

$$\bar{C}_{\text{out}} = \frac{\sum_{t \in \mathcal{T}} \sum_{i \in \mathcal{N}(t)} u_i(t)}{\sum_{t \in \mathcal{T}} |\mathcal{N}(t)|}. \tag{29}$$

After every rollout, the multiplier is updated by the projected step

$$\mu \leftarrow \Big[\mu + \eta_\mu \big(\bar{C}_{\text{out}} - C_{\text{thr}} \big) \Big]_0^{\mu_{\max}}, \tag{30}$$

and the policy is trained on r_μ. Because $\bar{C}_{\text{out}}$ is a rollout-level sample estimate of the constraint in (20), this procedure is a practical primal–dual surrogate of (21) rather than an exact solution. The multiplier is fixed within each rollout and frozen after training, does not modify the planner, and enters the decision through the Lagrangian reward and the feature $\mu/\mu_{\max}$ of the actor and the critic.

F. Training Procedure and Complexity

In each training iteration, all UEs collect a rollout of T_{roll} epochs with the current policy; the shared actor, the GRU, and the centralized critic are then updated with the clipped PPO objective [33], generalized advantage estimation [34], and truncated backpropagation through time on the team reward (r or r_μ), with early stopping at a target KL divergence and linearly annealed learning rates, and M5 finally updates μ by (30).

Execution complexity. Per epoch, a UE performs one recurrent update and one actor forward pass, and in M3–M5 the planner evaluates at most $\max\{2K+1, 4\}$ plans over $\tau_{\max}$ epochs, i.e., $\mathcal{O}(K\tau_{\max})$ per UE independently of N. On a single thread of an Intel Core i7-12700 CPU with an unoptimized Python/NumPy implementation (no GPU), the mean decision latency for all $N = 12$ UEs is 0.24–0.44 ms for the actor and at most 21.7 ms including the planner, i.e., about 1.8 ms per UE, far below the $\Delta = 1$ s epoch.

VI. Simulation Setup

A. Training

All variants use the environment and parameters of Table II. Each variant is trained for $I = 2000$ iterations with five independent training seeds $\{7, 8, 9, 10, 11\}$; the seed determines both the network initialization and the training-environment realization. The five M1–M5 variants therefore comprise 25 independently trained policies. Training episodes contain the randomized disturbance schedule of Section III; the compound-stress pattern is never used for training. To isolate goal regulation, we also train M1D, the memoryless M1 trained with r_μ and the dual controller of M5, with the same five seeds; the multiplier enters its critic input but not its actor input, although the actor is trained on r_μ.

B. Evaluation Protocol

Every trained policy is evaluated on five held-out evaluation seeds with 40 episodes each under the *nominal* and *compound-stress* conditions (Section III). All methods are evaluated on identical, mutually disjoint episode streams, which yields paired comparisons. Actions are sampled from the stochastic policies, prices are reset at the start of each episode, and the multiplier of M5 is frozen. For a learned variant, each training seed is first averaged over the evaluation seeds, and the mean and the standard deviation are computed across the five training seeds; differences between learned variants are tested with Welch's t-test on the per-seed means, and "significant" refers to the 5% level. Rule-based schemes are reported as means over the evaluation seeds. Besides the realized outage ratio,

we report the service deficit, the handover (HO) executions per epoch (summed over the N UEs), the ping-pong ratio (ping-pongs per successful handover), the resource cost C_{ctrl}, and the *common reward* r of (19), which is used for all methods including M5.

C. Baselines

All baselines observe the same delayed and noisy telemetry and are subject to the same action masks and admission rules as the learned agents.

- *CHO (A3/TTT)*: a 3GPP-inspired CHO that prepares the best observed candidate when it exceeds the serving satellite by 1 dB, executes it when it exceeds the serving satellite by an execution offset for a time-to-trigger (TTT), and releases it after it falls 1 dB below the serving satellite for three epochs. The offset (2, 3, 4 dB) and the TTT (1, 2, 3 epochs) were tuned for the common reward on three validation seeds disjoint from all evaluation episodes and frozen at 2 dB and one epoch, the most responsive values of the grid.
- *CHO (A3/TTT)+Dup*: the tuned CHO rule, additionally activating duplication while a target is prepared and the serving SINR is below a threshold tuned in the same way (6 dB, selected from $\{2, 4, 6\}$ dB).
- *Max-SINR*: prepares the best observed candidate when it exceeds the serving satellite by 1 dB and executes as soon as allowed, without duplication.
- *Always-Dup*: always prepares the best observed candidate, activates duplication, and executes.
- *Threshold-Dup*: prepares the best candidate when the serving SINR falls below 4 dB, activates duplication, and executes when the target exceeds the serving satellite by 1 dB.

Learning-based handover schemes from the literature [16]–[18], [20] select a serving satellite or trigger a handover, but they do not expose preparation, duplication, release, and rollback actions and therefore cannot be applied to the considered action space without redesign. We therefore use two learned baselines that operate on the same observation space, action space, action mask, and team reward r as the proposed agents and are trained with the same budget and seeds. *MAPPO (M1)* is the policy-based CTDE baseline using only the local perception input, i.e., without recurrent memory, ephemeris lookahead, congestion prices, or dual regulation [32]. *MADQN* (multi-agent deep Q-network) is a value-based baseline following the fingerprint approach of [18]: all UEs share one double deep Q-network ($[128, 128]$) whose input includes the training progress and exploration rate, trained with experience replay, a target network, and ϵ-greedy exploration ($1 \to 0.05$ over the first 40% of the iterations). Its learning rate was selected from $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ on the CHO validation seeds (10^{-3}), and it is evaluated with greedy actions, as is standard for value-based methods; to exclude an effect of this choice, M1, M1D, and M5 are additionally evaluated with argmax actions.

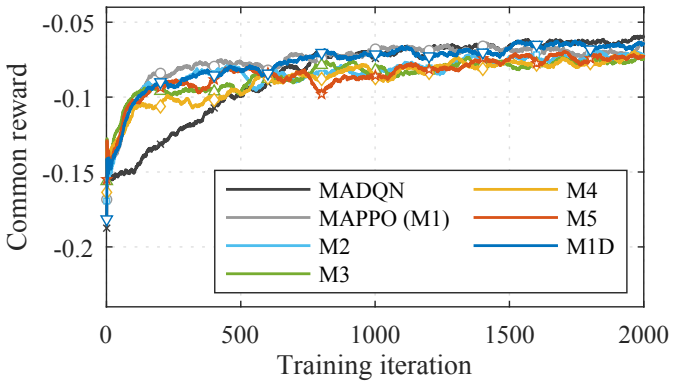


Fig. 3. Training convergence in terms of the common reward $r = -C_{\mathrm{def}} - wC_{\mathrm{ctrl}}$ (moving average over 50 iterations, mean over five training seeds). M5 and M1D are trained on the Lagrangian reward r_μ but are shown with the common reward for comparability.

VII. Results

A. Training Convergence

Fig. 3 shows the common reward during training. All variants improve from approximately -0.13 to -0.07 within the first few hundred iterations and then refine slowly under the annealed learning rates. Averaged over the last 100 iterations, M1 attains -0.067, M2, M4, and M5 about -0.072, and M3 -0.075, which mirrors the ranking on the held-out scenarios (Table IV). MADQN improves more slowly during its exploration phase but reaches the highest training common reward (-0.061). Quantitative comparisons use only the common held-out scenarios.

B. Comparison with Baselines

Table IV and Fig. 4 compare M5 with all five rule-based schemes and with the two learned baselines. All learned methods achieve lower outage ratios than the rule-based schemes: in the nominal condition, their outage ratios are 0.119–0.149, whereas the rules attain 0.322–0.429. Relative to the five rules, M5 lowers the outage ratio by 58–68% and the service deficit by 49–64%, raises the packet delivery ratio from 80.3–85.6% to 92.6%, and reduces the deadline-violation rate by 49–70% (2.2% versus 4.3–7.5%); under compound stress, the outage reduction is 44–56% (46% relative to CHO). M5 also has a 79–91% lower ping-pong ratio and 67–79% fewer failed handovers than CHO, CHO+Dup, Max-SINR, and Always-Dup. Threshold-Dup has lower mobility counts only because it rarely hands over, which yields the highest outage ratio.

Among the learned methods, the value-based MADQN attains the lowest mean outage ratio (0.119), a significantly lower service deficit than M5, and significantly fewer failed handovers than M1 and M5, whereas its outage difference to M1 and M5 is not significant; under compound stress, no outage difference among MADQN, M1, and M5 is significant, whereas M5 retains a slightly higher service deficit than M1 in both conditions (significant). M5, in contrast, has the lowest ping-pong ratio of all learned methods (0.035 versus 0.069 for MADQN and 0.118 for M1), a significantly lower preparation rejection rate than MADQN (20% versus 44%), and a stable

TABLE IV
PERFORMANCE UNDER NOMINAL AND COMPOUND-STRESS CONDITIONS. LEARNED MODELS: MEAN ± STANDARD DEVIATION ACROSS FIVE TRAINED POLICIES; RULE-BASED SCHEMES: MEAN OVER THE EVALUATION SEEDS. BOLD: BEST VALUE PER COLUMN AND CONDITION; UNDERLINED: BEST LEARNED VARIANT WHERE A RULE-BASED SCHEME IS BEST; SHADED: M5.

	Method	Outage ↓	Deficit ↓	HO exec./epoch ↓	Ping-pong ↓	Resource ($\times 10^{-3}$) ↓	Common reward ↑
Nominal	M1	0.131 ± 0.007	0.065 ± 0.004	0.367 ± 0.066	0.118 ± 0.022	5.48 ± 1.70	-0.066 ± 0.004
	M2	0.144 ± 0.017	0.072 ± 0.007	0.288 ± 0.054	0.060 ± 0.018	6.65 ± 3.04	-0.073 ± 0.008
	M3	0.149 ± 0.003	0.074 ± 0.001	0.317 ± 0.068	0.061 ± 0.014	6.24 ± 1.09	-0.076 ± 0.001
	M4	0.143 ± 0.011	0.071 ± 0.004	0.345 ± 0.092	0.075 ± 0.031	7.70 ± 3.89	-0.073 ± 0.005
	M5	0.137 ± 0.005	0.069 ± 0.002	0.224 ± 0.037	$\underline{0.035 \pm 0.006}$	$\underline{4.35 \pm 0.96}$	-0.070 ± 0.002
	MADQN	$\mathbf{0.119 \pm 0.016}$	$\mathbf{0.059 \pm 0.007}$	$\underline{0.209 \pm 0.063}$	0.069 ± 0.029	15.71 ± 17.72	$\mathbf{-0.062 \pm 0.011}$
	CHO (A3/TTT)	0.345	0.145	0.363	0.172	**2.64**	−0.146
	CHO (A3/TTT)+Dup	0.346	0.146	0.364	0.173	3.13	−0.147
	Max-SINR	0.322	0.137	0.515	0.337	3.30	−0.138
	Always-Dup	0.390	0.168	0.690	0.408	171.89	−0.211
	Threshold-Dup	0.429	0.190	**0.089**	**0.008**	9.01	−0.192
Stress	M1	0.194 ± 0.005	0.098 ± 0.003	0.384 ± 0.064	0.117 ± 0.023	5.75 ± 1.84	$\mathbf{-0.100 \pm 0.003}$
	M2	0.204 ± 0.016	0.105 ± 0.007	0.324 ± 0.063	0.066 ± 0.018	6.90 ± 2.84	-0.107 ± 0.008
	M3	0.208 ± 0.006	0.107 ± 0.002	0.352 ± 0.064	0.069 ± 0.010	6.48 ± 0.69	-0.109 ± 0.002
	M4	0.202 ± 0.008	0.104 ± 0.003	0.367 ± 0.094	0.075 ± 0.030	7.79 ± 3.90	-0.106 ± 0.004
	M5	0.199 ± 0.005	0.103 ± 0.003	0.255 ± 0.032	$\underline{0.040 \pm 0.007}$	$\underline{4.82 \pm 1.16}$	-0.105 ± 0.003
	MADQN	$\mathbf{0.189 \pm 0.020}$	$\mathbf{0.097 \pm 0.012}$	$\underline{0.233 \pm 0.071}$	0.075 ± 0.030	19.41 ± 20.76	-0.102 ± 0.016
	CHO (A3/TTT)	0.369	0.171	0.370	0.163	**2.68**	−0.172
	CHO (A3/TTT)+Dup	0.373	0.173	0.370	0.163	3.94	−0.173
	Max-SINR	0.352	0.164	0.531	0.328	3.38	−0.165
	Always-Dup	0.414	0.195	0.702	0.395	168.22	−0.237
	Threshold-Dup	0.457	0.220	**0.101**	**0.007**	17.64	−0.224

resource use, whereas two MADQN policies duplicate in 2.6% and 5.2% of the time. Across training seeds, the standard deviations of M5 are also 3.2, 1.7, and 5.2 times smaller than those of MADQN for the outage ratio, the handover executions, and the ping-pong ratio. These findings do not stem from the action-selection rule: with argmax actions, the outage ratios of M1, M1D, and M5 change by at most 0.5 percentage points (pp), whereas their handover executions and resource costs decrease by 10–21% and 36–41%, so that M5 attains the fewest handover executions (0.196), the lowest ping-pong ratio (0.031), and the lowest resource cost of all learned methods, while the outage advantage of MADQN persists and argmax M1 and MADQN are statistically indistinguishable in all metrics. A well-tuned value-based learner therefore provides a strong reliability baseline; M5 does not improve the outage ratio over MADQN or M1 but yields a lower ping-pong ratio and smaller across-seed variability.

The reliability gap relative to CHO also holds at the policy–scenario level: in each condition, all 125 policy–scenario pairs (25 trained policies $\times$ 5 evaluation seeds) have a lower outage ratio than CHO in the same scenario (250 comparisons in total), and the ratio between the outage of M5 and that of CHO stays within 0.38–0.42 (nominal) and 0.52–0.56 (stress) across the evaluation seeds. The learned agents also carry more traffic than CHO (utilization of about 50% versus 36%) at a lower peak requested pressure (1.13–1.20 versus 1.35). This reliability improvement uses more control resources than CHO, which almost never duplicates and thus spends 2.6×10^{-3} versus 4.3×10^{-3} for M5. A fixed SINR duplication trigger barely changes CHO (duplication active in 0.06% of the time). Under compound stress, whose burst and blockage severities lie outside the training range, all learned methods remain far more reliable than the rules, but their outage ratio increases by 5.9–7.0 pp, compared with 2.5 pp for CHO and 3.0 pp for Max-SINR, i.e., they are more sensitive to out-of-distribution disturbances.

C. Capability-Structured Comparison

Fig. 5 and Table IV show that the capability ladder is not monotone and that the capabilities mainly change *how* reliability is achieved. The memoryless M1 attains the lowest outage ratio among M1–M5 and the best common reward, but with the most mobility actions. *Memory* (M2) halves the ping-pong ratio and reduces handover executions by 21% at a 1.3 pp higher outage ratio, which is consistent with fewer reactions to transient telemetry fluctuations. *Reasoning* (M3) yields no measurable gain over M2, and *coordination* alone (M4) gives mixed results. The largest change occurs when *goal regulation* is added (M4→M5): handover executions decrease by 35% and the ping-pong ratio by 53% (both significant), and the standard deviation across trained policies shrinks from 0.092 to 0.037 for the handover executions and from 0.031 to 0.006 for the ping-pong ratio. Compared with M1, M5 performs 39% fewer handover executions and has a 70% lower ping-pong ratio (both significant) at a non-significant 0.6 pp outage difference and a significant 0.5 pp higher service deficit; under compound stress, the reductions are 34% and 66%. M1 and M5 therefore form two distinct operating points.

To isolate goal regulation, M1D applies the Lagrangian reward to the memoryless policy. Compared with M1, M1D lowers the ping-pong ratio by 38% (significant) and the handover executions by 28% (not significant) at a statistically indistinguishable outage ratio (0.126) and common reward, and all 50 M1D policy–scenario pairs (25 per condition)

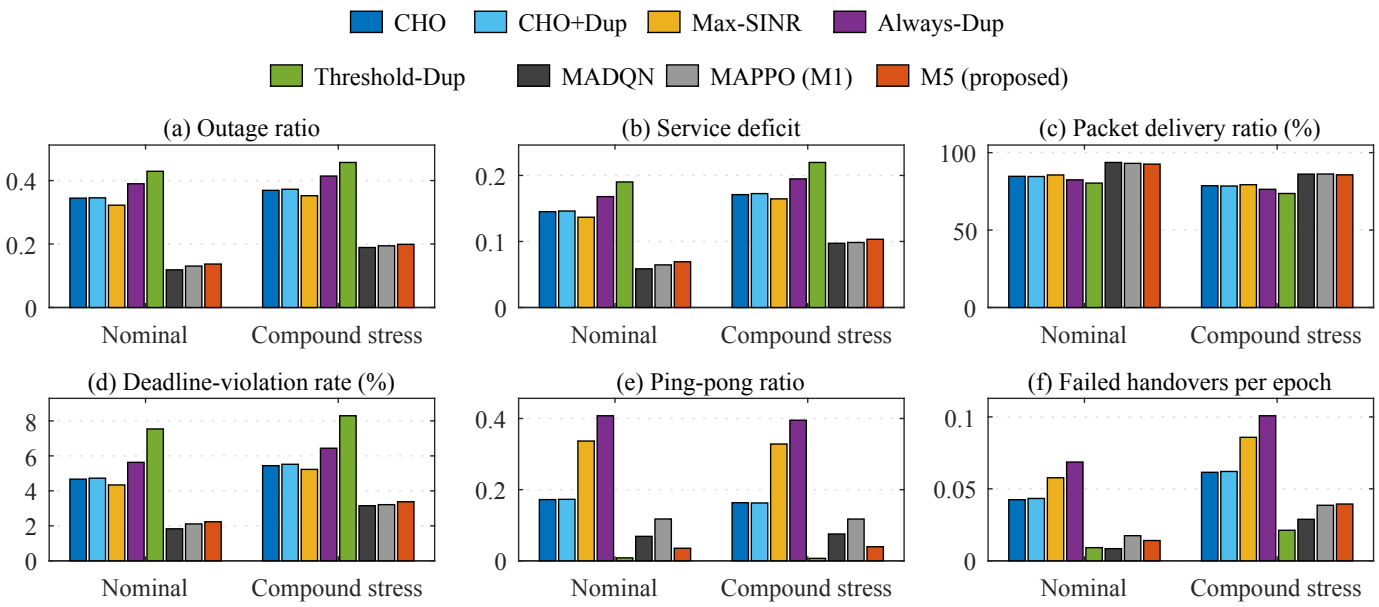


Fig. 4. Comparison of M5 with the five rule-based schemes and with the MADQN and MAPPO (M1) baselines under nominal and compound-stress conditions (means over five trained policies for learned methods and over five evaluation seeds for rule-based schemes; standard deviations of the learned methods are given in Table IV). Threshold-Dup attains low ping-pong and handover-failure counts only because it rarely hands over.

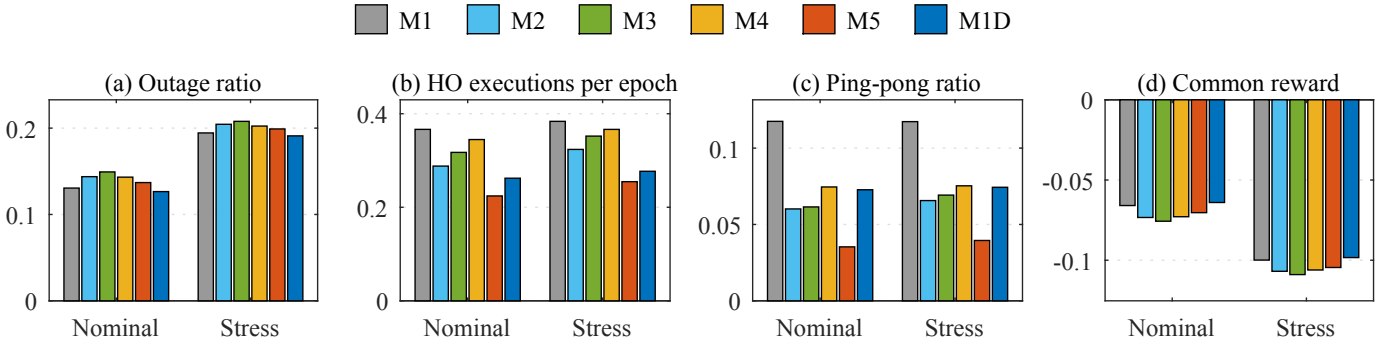


Fig. 5. Capability ablation: the model family M1–M5 and the goal-regulation-only variant M1D under nominal and compound-stress conditions (means over five trained policies).

again outperform CHO. Goal regulation therefore produces the largest isolated reduction in handover executions among the tested additions, whereas memory alone yields the larger ping-pong reduction (49% versus 38%). The remaining capabilities still change the operating point: relative to M1D, M5 halves the ping-pong ratio again (-51%, significant) with markedly smaller standard deviations across trained policies, but has a significantly higher outage ratio ($+1.0$ pp), a lower common reward, and 24% more preparation requests. Post hoc, M5 overtakes M1 in common reward only when handover executions are weighted about 30–35 times more strongly than in training, whereas M1D exceeds M5 for all tested weights. Memory, reasoning, and coordination thus trade a small amount of reliability for further ping-pong suppression and training consistency.

D. Reliance on Coordination and Reasoning Signals

To test whether the trained policies rely on the price and on the planner, we re-evaluate them on the same episodes with the price forced to zero (M4, M5) or with an uninformative reasoning feature (M3–M5), i.e., $\phi_{i,a} = 0$ for feasible and $\phi_{i,a} = 1$ for masked actions, the value observed during

TABLE V
WITHIN-POLICY ABLATIONS: RELATIVE CHANGE WHEN THE CONGESTION PRICE IS FORCED TO ZERO OR THE REASONING FEATURE IS MADE UNINFORMATIVE AT EVALUATION (MEAN OVER FIVE TRAINED POLICIES), WITH THE NUMBER OF DEGRADED POLICIES OUT OF FIVE IN PARENTHESES. *: SIGNIFICANT CHANGE (PAIRED t-TEST, 5% LEVEL).

		Price removed		Reasoning removed		
	Metric	M4	M5	M3	M4	M5
Nominal	HO exec.	+18.5* (5)	+23.6* (5)	−1.1* (0)	+6.9* (5)	+8.0* (5)
	Ping-pong	+35.4* (5)	+21.6* (5)	−3.1 (1)	+10.2 (5)	+8.4* (5)
	Resource	+24.4 (5)	+22.6* (5)	−2.2 (1)	+6.0 (5)	+5.4* (5)
	Outage	+6.9* (5)	+5.3* (5)	+0.9 (4)	+3.6* (5)	+3.1 (4)
	Com. reward	−6.1* (5)	−4.0* (5)	−0.7 (3)	−3.3* (5)	−2.8* (5)
Stress	HO exec.	+17.9* (5)	+20.4* (5)	−0.3 (3)	+6.2* (5)	+6.2* (5)
	Ping-pong	+33.5* (5)	+26.4* (5)	−2.2* (0)	+10.1* (5)	+8.7* (5)
	Resource	+26.3* (5)	+20.3* (5)	−0.0 (3)	+5.3 (5)	+3.1* (5)
	Outage	+2.5 (4)	+1.6 (4)	+0.2 (3)	+1.6 (4)	+1.2 (4)
	Com. reward	−2.3 (4)	−1.4 (4)	−0.2 (2)	−1.6 (4)	−1.0 (4)

training whenever all plan costs are tied (Table V; changes in %). A policy is counted as degraded when its handover executions, ping-pong ratio, resource cost, or outage ratio increase, or when its common reward decreases. Removing the price increases the handover executions by 18–24%, the ping-pong ratio by 22–35%, and the resource cost by 20–26% in all ten M4/M5 policies and in both conditions; the outage ratio increases in all ten policies in the nominal condition (+5–7%) and in eight of ten under stress, where the change is small and not significant. M3 shows little sensitivity to the removal of the reasoning feature: its changes remain within about 3% without a consistent direction. In M4 and M5, however, the reasoning feature also contains the price-aware plan cost of (25); removing it increases the handover executions by 6–8%, the ping-pong ratio by 8–10%, and the resource use by 3–6% across all ten policies and both conditions. The reasoning input therefore affects the trained policies primarily when it is coupled with congestion information. Because removing a trained input creates a distribution shift, these ablations measure within-policy reliance rather than the causal benefit of training with the signal.

E. Conditional-Handover Behavior

Fig. 6 shows that the learned agents use the CHO procedure differently from the rules. Once the tuned CHO rule prepares a target, it executes the switch in 73% of the cases after holding the preparation for 5.3 epochs, and 49% of its requests are rejected because many UEs request the same satellite at the same time. The learned agents instead treat a preparation as a short-lived *option*. M5 issues a preparation in 38% of its decisions and releases most of them, so that only 6.3% of the preparations end in a switch, and it holds a preparation for only 1.3 epochs. Across M1–M5, the prepare-to-commit ratio decreases from 18.9% for M1 to 9.7–13.5% for M2–M4 and 6.3% for M5, while the preparation rejection rate decreases from 35% to 20%. PDCP duplication is used sparingly by the MAPPO-based variants (active 0.2–0.6% of the time), whereas activating it indiscriminately (Always-Dup) degrades every metric. This behavior increases control-plane signaling: M5 issues 4.6 preparation requests per epoch, compared with 1.0 for CHO, which the present reward does not penalize (Section VIII).

VIII. Discussion and Limitations

Reliability target. The outage ratios of M5 and M1D on the held-out scenarios (0.137 and 0.126) remain above the threshold $C_{\mathrm{thr}} = 0.10$, as do those of all other variants, and changing the frozen multiplier at evaluation hardly changes the behavior of M5. The dual controller should therefore be understood as a training-time regulator rather than as a goal-conditioned policy; enforcing the constraint on unseen scenarios and training with randomized multipliers are open problems.

Reasoning. M3 shows little behavioral dependence on the purely geometric planner, whereas the price-aware planner of M4 and M5 affects the trained policies. The planner extrapolates delayed telemetry, treats a feasible switch as deterministic, and sees control-slot contention only through the price; an expected handover-success model, an explicit contention term, and planner-based action priors are natural next steps. Why the Lagrangian reward alone suppresses unnecessary handovers (M1D), and why M5 further reduces ping-pong and training variability, also remain open questions.

Signaling and duplication. The reward charges reserved preparation resources but not preparation messages, so the learned agents issue 3.6–4.9 times more preparation requests than CHO; a message-level signaling cost is needed to confirm that the advantages persist when signaling is priced. The regimes in which duplication becomes worthwhile (e.g., stricter reliability classes or lower duplication cost) also remain to be characterized.

Model scope. The study uses a stylized slice with six satellites and twelve UEs, a fixed interference floor, a system-level protocol abstraction, a fixed scheduler, and five trained policies per variant. Larger constellations, explicit interference, and message-level protocol effects remain to be evaluated.

IX. Conclusion

We formulated the joint control of CHO preparation, PDCP duplication, and primary-satellite switching in a shared-capacity LEO network as a constrained Dec-POMDP and developed a capability-structured multi-agent framework with memory, ephemeris-based lookahead, congestion-price coordination, and training-time reliability regulation. Every learned policy–scenario pair achieved a lower outage ratio than the tuned 3GPP-inspired CHO rule, and M5 reduced the outage ratio by 60% under nominal conditions and by 46% under compound stress relative to CHO. Compared with the memoryless MAPPO policy, M5 attained a statistically indistinguishable outage ratio with 39% fewer handover executions and a 70% lower ping-pong ratio. A tuned multi-agent DQN provided a strong reliability baseline, while M5 produced a lower ping-pong ratio and smaller across-seed variability in mobility behavior. Ablations showed that goal regulation produced the largest isolated reduction in handover executions, that the M4 and M5 policies relied on congestion prices, and that the reasoning feature affected behavior mainly when its plan cost included the price term. The learned policies used CHO preparations primarily as short-lived reservations. Given the limitations of Section VIII, future work should evaluate goal-conditioned regulation, planner-guided action selection, and signaling-aware rewards.

References

[1] M. Giordani and M. Zorzi, "Non-Terrestrial Networks in the 6G Era: Challenges and Opportunities," *IEEE Netw.*, vol. 35, no. 2, pp. 244–251, 2021.

[2] 3GPP, "Study on New Radio (NR) to Support Non-Terrestrial Networks," TR 38.811 V15.4.0, 2020.

[3] G. Fontanesi *et al.*, "Artificial Intelligence for Satellite Communication: A Survey," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 1381–1435, 2026.

[4] E. Juan, M. Lauridsen, J. Wigard, and P. Mogensen, "Handover Solutions for 5G Low-Earth Orbit Satellite Networks," *IEEE Access*, vol. 10, pp. 93309–93325, 2022.

[5] S. Ji, M. Sheng, D. Zhou, W. Bai, Q. Cao, and J. Li, "Flexible and Distributed Mobility Management for Integrated Terrestrial-Satellite Networks: Challenges, Architectures, and Approaches," *IEEE Netw.*, vol. 35, no. 4, pp. 73–81, 2021.

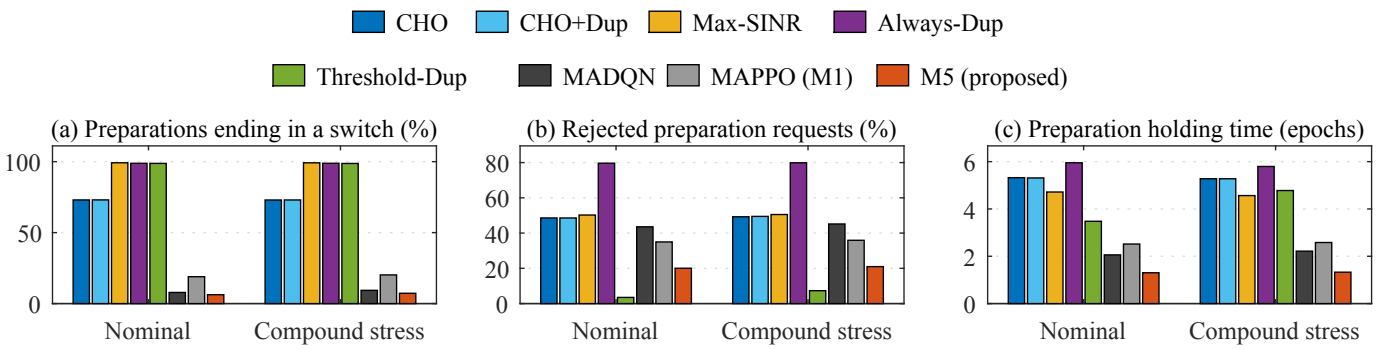


Fig. 6. Use of CHO preparations by the rule-based schemes, the learned baselines, and M5 under nominal and compound-stress conditions.

[6] S. Zhang, A. Liu, C. Han, X. Ding, and X. Liang, "A Network-Flows-Based Satellite Handover Strategy for LEO Satellite Networks," *IEEE Wireless Commun. Lett.*, vol. 10, no. 12, pp. 2669–2673, 2021.

[7] J. Lee, W. Lee, and J.-H. Kim, "Performance Evaluation of Location-Based Conditional Handover Scheme Using LEO Satellites," in *Proc. International Conference on Information and Communication Technology Convergence (ICTC)*, 2023, pp. 1642–1644.

[8] J. Wu, S. Su, X. Wang, J. Zhang, and Y. Gao, "Accelerating Handover in Mobile Satellite Network," in *Proc. IEEE INFOCOM*, 2024, pp. 531–540.

[9] J. Li *et al.*, "SKYCASTLE: Taming LEO Mobility to Facilitate Seamless and Low-Latency Satellite Internet Services," in *Proc. IEEE INFOCOM*, 2024, pp. 541–550.

[10] S. Wang, Z. Jiang, X. Li, and J. Ma, "PMPS: Predictive Multi-Path Scheduling for Handover-Free LEO Communications," in *Proc. IEEE INFOCOM*, 2026, pp. 1–10.

[11] C. Pupiales, D. Laselva, Q. De Coninck, A. Jain, and I. Demirkol, "Multi-Connectivity in Mobile Networks: Challenges and Benefits," *IEEE Commun. Mag.*, vol. 59, no. 11, pp. 116–122, 2021.

[12] M. Lopez, S. B. Damsgaard, I. Rodriguez, and P. Mogensen, "An Empirical Analysis of Multi-Connectivity Between 5G Terrestrial and LEO Satellite Networks," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, 2022, pp. 1115–1120.

[13] 3GPP, "Solutions for NR to Support Non-Terrestrial Networks (NTN)," TR 38.821 V16.2.0, 2023.

[14] 3GPP, "NR; Radio Resource Control (RRC) Protocol Specification," TS 38.331 V19.1.0, 2025.

[15] 3GPP, "NR; Packet Data Convergence Protocol (PDCP) Specification," TS 38.323 V19.1.0, 2025.

[16] J. Yang, Z. Xiao, H. Cui, J. Zhao, G. Jiang, and Z. Han, "DQN-ALrM-Based Intelligent Handover Method for Satellite-Ground Integrated Network," *IEEE Trans. Cogn. Commun. Netw.*, vol. 9, no. 4, pp. 977–990, 2023.

[17] H. Liu, Y. Wang, P. Li, and J. Cheng, "A Multi-Agent Deep Reinforcement Learning-Based Handover Scheme for Mega-Constellation Under Dynamic Propagation Conditions," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 13579–13596, 2024.

[18] F. Yang, W. Wu, Y. Gao, Y. Sun, T. Sun, and P. Si, "Multi-Agent Fingerprints-Enhanced Distributed Intelligent Handover Algorithm in LEO Satellite Networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 10, pp. 15255–15269, 2024.

[19] F. Wang, H. Yang, S. Zhang, X. He, L. Zhu, and T. Q. S. Quek, "GEMS: GAT-Embedded Mobility for Dual Connectivity in Integrated Terrestrial and Non-Terrestrial Networks," *IEEE Trans. Mobile Comput.*, vol. 25, no. 10, pp. 15643–15659, 2026.

[20] D. Zhao, Y. Wang, B. Song, Y. Zhou, and P. Qin, "Learning When and Where to Handover: A Hierarchical Reinforcement Learning Framework for Dense LEO Satellite Constellations," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 12787–12801, 2026.

[21] R. Sapkota, K. I. Roumeliotis, and M. Karkee, "AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges," *Inf. Fusion*, vol. 126, Art. no. 103599, 2026.

[22] F. Jiang, C. Pan, K. Wang, P. Michiardi, O. A. Dobre, and M. Debbah, "From Large AI Models to Agentic AI: A Tutorial on Future Intelligent Communications," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 3507–3540, 2026.

[23] R. Zhang *et al.*, "Toward Edge General Intelligence With Agentic AI and Agentification: Concepts, Technologies, and Future Directions," *IEEE Commun. Surveys Tuts.*, vol. 28, pp. 4285–4318, 2026.

[24] J. Zheng *et al.*, "Advanced Deep Reinforcement Learning for Agentic AI and Their Applications in Wireless Network," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 7522–7544, 2026.

[25] Z. Ni, Z. Hua, X. Yang, R. Zhang, S. Wang, and G. Pan, "LLM-Aided Spectrum-Sharing LEO Satellite Communications," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 2591–2605, 2026.

[26] J. Shen *et al.*, "Agentic AI for 6G Satellite Networks: Achieving Scalable and Low-Latency Resource Scheduling With the AERS Framework," *IEEE Netw.*, early access, 2026, doi: 10.1109/MNET.2026.3692763.

[27] Y. Xie, Q. Song, W. Sun, L. Guo, and A. Jamalipour, "Task-Oriented Agentic AI-Driven Resource Orchestration in Satellite-Assisted IoV," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 10559–10572, 2026.

[28] H. Cao *et al.*, "A Think-Do Architecture for Cognitive Autonomy: LLM-Driven Agentic AI for Sustainable Resource Orchestration in 6G Satellite Networks," *IEEE Netw.*, early access, 2026, doi: 10.1109/MNET.2026.3699359.

[29] L. Zhang, H. Zhao, B. Xu, H. Zhu, and X. Wang, "Agentic AI for SAGIN Resource Management: Semantic Awareness, Orchestration, and Optimization," *IEEE Netw.*, early access, 2026, doi: 10.1109/MNET.2026.3695611.

[30] C. Hu, H. Yang, B. Li, X. Jiang, and W. Lu, "LLM-Empowered MAPPO for Embodied Cognitive Satellite-Terrestrial Networks With RSMA," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 10855–10870, 2026.

[31] M. Zhang, Y. Sun, X. Liu, and H. Zhou, "Robust and Stable Multipath Transmission in Large-Scale LEO Satellite Networks: An Agentic AI-Driven Approach," *IEEE Trans. Cogn. Commun. Netw.*, vol. 12, pp. 11102–11115, 2026.

[32] C. Yu *et al.*, "The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games," in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 24611–24624.

[33] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," arXiv:1707.06347, 2017.

[34] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-Dimensional Continuous Control Using Generalized Advantage Estimation," in *Proc. ICLR*, 2016.